

Optimized Return Logistics for Quick Commerce

Implementing Weighted Cost Functions for Fraud Prevention and Route Optimization

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Abstract

This document outlines a solution to address refund fraud in Quick Commerce applications (e.g., Swiggy Instamart, Zepto). We propose a physical verification model where items are retrieved by delivery partners. To optimize this process, we utilize a **Weighted Cost Function** that integrates distance (space) and traffic (time/friction) to determine the most efficient routing for returns.

1 Problem Statement

In the current Quick Commerce ecosystem, transparency regarding refunds is a significant challenge.

1. **Verification Gap:** Users may claim a product is damaged or unhygienic, but the uploaded photographic evidence can be misleading.
2. **AI Manipulation:** With the rise of Generative AI, photos can be edited or generated (e.g., deepfakes) to simulate damage, leading to fraudulent refund claims.
3. **Manufacturer Accountability:** For packaged foods, defects often originate from the manufacturer, not the delivery service. Physical retrieval is necessary to prove liability.

2 Proposed Solution: Physical Retrieval Loop

To mitigate fraud (potentially verified via tools like SynthID) and ensure accountability, we propose a reverse-logistics model:

- A Delivery Partner is dispatched to the Consumer.
- The item is retrieved and transported to a Warehouse for verification.
- The refund is processed only upon physical receipt and verification at the warehouse.

3 The Mathematical Model: Weighted Cost Function

To ensure this process does not incur excessive operational costs, we must select the optimal route. We convert two distinct variables—Distance (D) and Traffic Intensity (α)—into a single scalar value called the **Effective Score**.

3.1 The Core Equation

The cost of traversing a path is defined as:

$$\text{Effective Score}(E) = D \times (1 + \alpha) \quad (1)$$

Where:

- D : Physical Distance in kilometers.
- α : Traffic Factor (Percentage converted to decimal, e.g., 40% = 0.40).
- $(1 + \alpha)$: The Multiplier Effect. A value of 1.0 represents a clear road; values > 1.0 represent friction.

3.2 Algorithm Logic

The algorithm performs two distinct steps to minimize total effort:

Step 1: Partner Selection (Minimize Arrival Time)

Identify the Delivery Partner closest to the Consumer (C_o).

$$\min(D_{\text{Partner} \rightarrow C_o})$$

Step 2: Warehouse Selection (Minimize Return Cost)

Evaluate all available Warehouses (W_i) based on the Weighted Cost from the Consumer.

$$\text{Optimize: } \min(D_{C_o \rightarrow W_i} \times (1 + \text{Traffic}_{C_o \rightarrow W_i}))$$

4 Numerical Analysis (Case Study)

Let us analyze the scenario provided in the system specifications.

4.1 Inputs

1. Delivery Partners (Distance to Consumer):

- Partner 1: 1 km
- Partner 2: 5 km
- Partner 3: 3 km

2. Warehouses (Distance and Traffic from Consumer):

- Warehouse A: 11 km distance, 40% traffic.
- Warehouse B: 12 km distance, 11% traffic.

4.2 Step 1: Select Delivery Partner

Comparing the distances:

$$1 \text{ km} < 3 \text{ km} < 5 \text{ km}$$

Selected: Delivery Partner 1 (1 km away).

4.3 Step 2: Select Warehouse (The "Trap" vs. The "Smart Choice")

Option A (Warehouse A):

$$E_A = 11 \times (1 + 0.40) = 11 \times 1.40 = \mathbf{15.4}$$

Interpretation: The high friction makes the 11km trip effectively cost as much as 15.4km.

Option B (Warehouse B):

$$E_B = 12 \times (1 + 0.11) = 12 \times 1.11 = \mathbf{13.32}$$

Interpretation: Despite being physically longer, the low friction results in a lower effective cost.

4.4 Conclusion

Since $13.32 < 15.4$, the algorithm routes the partner to **Warehouse B**. This demonstrates that distance alone is an insufficient metric for logistics in high-density urban environments.